

The Emotional Impact of Covid-19 News.mp4 transcript with timestamps

00:00

Hi there! We are starting our presentation called Interpreting Tweets in the Media World in COVID-19 Pandemic. My name is Francisco. I am a data scientist. I graduated in electrical engineering in Brazil. I am a master candidate at USP, also in Brazil. And I have more than 10 years with experience using statistics, machine learning, artificial intelligence in both academic and industrial fields. The content of this presentation have four main topics, which are NLP, natural language processing, text processing, extracting topic sentiments and emotions, and in the last part we are going to talk about one paper that we published that used all those information to find the role of the media during the COVID-19 pandemic.

01:07

The first topic NLP, Natural Language Processing. The first question that we might ask is what do we want? What do we expect from NLP? We expect machines capable of mapping the constructions of one or several languages, or machines that interpret and understand human language. As you can see, those two concepts might be similar, but they are very different. One thing is to map the constructions of one language or several languages, how the words are related and how do you build one phrase, one sentence. And the other thing is to truly understand human language. But maybe to answer this question, we can first answer the other question, which is what products do we want NLP to offer us? I have here some products, some things that we use in our daily basis with NLP.

02:23

The first one, Google, all of you guys know it, but one thing that Google can do using NLP is to fix one text. For example, I am asking Google what is the temperature of the sun, but I misspelled a lot of words. And Google fixed that for me. This is an NLP application that does not need to fully understand one language. It just needs to know how words are positioned in a phrase, which is the first option. I also can use Google to translate text. I also can use some tools to fix the grammar of text. And all those options do not need to understand the language. They just need to know how the grammar works, how the syntax works, how the morphology works. There is another application which you guys might be familiar with, which is the chat GPT. And you can

n use it, for example, to ask how do you do a hello world in different programming languages? So here we have Java, Python, JavaScript, C++, PHP. And that's also, if we think one programming language as a language, it also has a structure. You can map this structure to build something.

04:09

But if you are trying to use a virtual assistant, if any of you guys have used a virtual assistant someday, you might find it very difficult to communicate with them. It's much more difficult than to communicate with a human being. This is because those models do not fully understand the language. JetGPT has very convincing answers, but many times if you use a lot, you can see that sometimes they give you wrong answers or answers that do not make any sense. And there is one nice metaphor that we get away from the text you can pause the video and read the text but the idea of this metaphor is that there is one guy inside of a box and that guy do not understand English oh sorry do not understand Chinese. And there's one Chinese guy putting questions here. This guy that do not understand Chinese reads a book with Chinese rules of how the grammar works or how to phrase answers. And with the help of the book, he outputs one answer to other guy. If this guy is very good or the book, he outputs one answer to other guy. If this guy is very good, or the book is very good, the answers would be understandable to the second guy. And the guy might think that this box truly understands Chinese, but it does not. If the input is something different, something new, something that is challenging, the answer will be not good. This is the point. This is more or less how artificial intelligence models works nowadays. They do not truly understand the language, but they are very good to understand the rules of the language and give nice answers. So we need to be aware of that. And also there are many, many, many challenges nowadays still to truly understand one language. For example, how do one model interpret sarcasm? It's very hard. How do one model detect patronizing and condescending language? It's very hard. How do one model works not only in English but in multilingual scenario? It's very hard. So there are many, many challenges still today and all those challenges here, there are a lot of researchers working on that to make models better. So although our models are very good today, there are a long way to improve.

07:35

So now we can go to the text processing tweets. As you can imagine, if anyone here

have ever used Twitter, many people don't write in Twitter like what you find in books. Many times they write very wrongly or using slangs or etc. So for example, I wonder if John Wren is kicking himself for not getting the COVID-19 vaccine. This text is pretty neat, but there's one problem. The name of the guy is with a hashtag. So if you are going to process this text using NLP, If you are going to process this text using NLP, that is a problem because many models don't understand that this is a name of a person because it's using, but the content of the tweet from a news venue, right? So COVID-19, New Bellevue, I guess this is from Canada, Ontario, maybe. Vaccine clinic opens at Queen's Sports and Wellness Center. And there is a red link here, which is also a problem if you are going to use NLP. Another one, this is BS. Yes, you guys know what it means. There is no COVID in SA, it stands for South Africa. So BS. I know you guys know what that means, but one model doesn't know, right? The last one. Oh, thanks, friend. This all is also hard to interpret for a model. Dot, dot, dot. It's also a problem. How to go clean out the COVID error. and then there's an emoji of a car. It's also a problem for our model. Transmission crapped out on me yesterday. Then I got caught up in some music, another emoji, another problem. So as you guys can see if you are going to implement a model here to for instance try to find which is the emotion the sentiments which uh of each of these tweets that will be a problem because uh many tweets have problems in the text, as we saw. And for not fixing all the errors, but reducing the amount of problems in each text, we can use this library in Python called Spacey, which is pretty interesting. It helps us a lot. And now we can make a small demonstration here of how it works. So that's going to be a live demonstration. And I hope that can be useful for you guys to know a little bit of how the code works in a project like that.

11:08

So we are going to use pandas. Okay and I am using that code here to take a sample of tweets that we have on our data set. That sample, as you can see, it shows the tweet ID, the conversation ID, it's a very long number, the author ID, that number relates to the user, the Twitter user who tweeted it. In reply to user ID, so all those tweets that I'm showing you guys are replies to news in Twitter. So they are always replying, maybe in that case no, but they are mostly replying to another account, another Twitter account, which is represented here by this number. That is was created, the geolocation, if available, the language, number of likes, number of quotes, number of replies, of retweets, and also this one which is what we are looking for the text right so we can

n see the text is like that or we can put those texts inside of inside of a list so we can see the whole content of each tweet like this one another one dot will be half the cabinet of at this rate maybe someone was fired right uh so those are the treats uh for example this one is replying to sky news that's why sky news is quoted in the beginning, right? If I can see one example of this list, I can use the practice here. Not that I was ever going to go to a Eric Clapton concert anyway, but I definitely wouldn't go now on principal. Nice reply.

13:51

Okay, so we can start using space, import space, right? And I have saved here a code where we can download what we call here a model. Right? This model is the English Core Web Medium. So it's a model that Space is trained on. So it's a model that Space is trained on, and it's used for Space to make all the things that we're going to do next. So, for example, let's see one tweet. Let's see tweet number 500. Or number 100, the same tweets that we saw earlier we want to convert this to this tweet to a document the document is the unit that we can work on using space right so we are going to use an mp to it So we are going to use an LP to read. And now we can go doing a for loop for each token in a document. The token is the unit of a document. A document is a collection of tokens. So if I say I love you, I love you is the document, I is one token, love is another token, you is another token. So for instance, if I print here all the tokens text, tokens text this is going to be the result not that I was ever going but you can see that comma is one token and the web link is another token right I can also use spacing to check a few things like if the token is punct, punct is punctuation. Let's see which of those are punctuations. False, false, false, false. The only true is the comma. Right. Also I can using the token to check if the token is alphanumeric false would be true and T is false if you if it was then there was not like the first one here it would be true and also the web link is also false there's one interesting thing that we can use which is int type. For instance this model, there are a few models I'm using core web medium, there's core web small, core large, and each one is more accurate than the previous one, right?

17:27

So this one, ops, I can use the underline. So I'm seeing that Eric Clapton, Eric is a person, Clapton is a person, and the other words are nothing like that right so detecting things like that is based on models and the model that is detecting is this one right s

o i have a small uh for us to not lose much time here. I'm reproducing this code. What this code does, it prints the document before and prints the document after. Before and after what? A pre-processing step. So I want to, for instance, I want to replace any token that is labeled as person with the word person is the last word, I'm not doubling it. So for example, Eric and Clapton, those two are persons, right? If I replace each one with the word person, it would be person. So I just keep that first. And I'm in, I am also, uh, removing or not including any token that is, it's like a URL in the world that is like a punctuation in the world that starts with, uh, at or with hashtag. So if we look at the text before this processing step, I removed this first part, I removed the comma, and I replaced Eric Clapton with the word person. Not that I was ever going to go to a person concert anyway, but I definitely would not go now on principle. So it's different. We can try to use maybe another tweet and see how that code works. Which South Africa? And there is a web link. The text afterwards is which South Africa without the interrogation mark and without the web link. Maybe another one. ENCA, this guy is a bit slow, like by about 20 years, dot, dot, dot. This guy is a bit slow, like by about 20 years. So it's cleaner now cleaner now right so we can use this cleaner text to analyze and find emotional sentiments in something like that okay uh i guess this is a small introduction and there are many things we can do with some code like that.

02:06

But for now, let's go back to our presentation. So this is a space. It can, as I showed you, it can find if our token is, is the alpha numeric token. It shows the shape, like the first one is capitalized, is a capitalized layer in the order of null, it shows the morphological pattern, like it's a verb, noun, something like that, it's a number, and many, many other things. So, one NLP model is able to, for instance, for example, find the similarity of two words, and how it does that. It represents each word in any- space. So each word is converted in a vector of numbers, a vector for, with, for example, 300 positions and numbers from zero to one. And you can find the Euclidean distance of those two words represented in vectors using Euclidean distance. And using that you can find if two words are similar or not. And it's interesting that if you try to represent that 300 dimensional vector in three dimensions. You can, for example, see that the distance from man to woman is similar with the distance from king to queen, or walking to walk is similar to swimming and so on. As you can see, and countries and capitals are the same. So it's interesting to learn that the model can find the model that cre

ates those vectors that represents each world. It represents those worlds in a space that makes sense. That words that are similar are close and words that have the same difference have the same distance. With that, we can rely on those kinds of models and NLP models, to do things more elaborated, like, for instance, finding the emotion, finding the sentiment of one text, which is our next topic, extracting topic sentiments and emotions from a dataset.

24:06

Topic modeling is the task of extracting latent topics from a set of documents automatically. So imagine that you have a hundred news headlines and you want to find which are the topics that those headlines are talking about. You don't know the topics previously, you just have the news headlines and you want a model to find which are the latent topics, meaning the topics that can be observed there. So if you can do that with a model, that is called topic modeling. The second one is sentiment and emotion analysis, which is the task of classifying each document, so each text, each tweet, for example, into a set of sentiments and or emotions. For example, imagine that those a thousand news headlines that I talked about in the previously the previously in the previous slide are all about COVID-19. And after running a topic model, you find that those are the word clouds, the three main word clouds that you found on those news headlines. So, for example, the first word cloud have the words, the most strong words, school, social, office, students, distance, staff, family, stay, century, medical, essential. The second one is economy, dollar, minister, market, cent, impact, price, company. And the third one is quarantine, port, flight, world, in fact, ship, infection, Wuhan, patient. patient. So as you can see, this is the result of a LDA topic model, which is one kind of technique that you can use to perform topic modeling, which is the Latent Dirichlet Allocation.

02:39

That topic model does not give you the name of the topic, right? So remember, you have 1,000 news headlines about COVID-19 and you run a topic model over those new headlines. What the topic model outputs to you is a word cloud, for example, of the main topics, and you have to say how many topics you want. So for example, I say I want three topics and the word clouds that the model gave me are those. And I have to look through those word clouds and name the topics. For example, clouds and name the topics. For example, maybe their first topic would be prevention, social im

pact of COVID-19, something like that. would be preventive measures, maybe the spread of the disease, something like that. So the model does not give you everything, but it does give you a nice hint of what is going on those set of documents. And this is another technique that you can use to talk modeling, which is top to vec, which instead of taking the documents like LDA, it takes the vectors, right? the transformation of one document in vector and finds that rather where in this n-dimensional space and where there is a more area, right? So, for instance, this area here, this light yellow area here, is more dense than average and the model says that this is one topic and you can see the documents that are inside of this topic and find a way to name it. But in this case, there are many, many documents that are not in any topic, which are the brown, brown, red, something like that points in this image. So this is another technique and there's a red link here that you can use to So this is another technique. And there's a web link here that you can use to understand a little deeper how it works. But those are, I would say, more used, more classic techniques that you can use top key modeling, LDA and topic2vec.

29:47

And for the task of sentiment analysis, which is when I want to find the sentiment, positive or negative, of a text. And even with the emotional analysis, when I want to find the emotion of one's document, like angry, sad, better than that, anger, sadness, joy, something like that. joy, something like that. For those kind of tasks, I can use something a little more robust called BERT, which is an NLP model based on... It's like a machine understanding how the construction of the language works, right? Remember one of the first slides I talked about that birth works like that. And and you can fine tune, you can adjust, adjust this model to tasks like sentiment analysis and emotional analysis. So birth is a name that maybe you might keep in mind and read a little bit more about it.

31:10

Okay, so I was not too far in understanding how sentiment analysis works, topic modeling works, but I'm going to show you some results of applying those techniques in a real data set, which is the role of the media during the COVID-19 pandemic, which is a paper that me, Dr. Davoud, and many other smart researchers wrote, I guess, one year ago. We used 1.7 million news tweets from 276 media accounts, spanning 12 countries across all five continents. All those countries, English speaking countries, a

and we had media, Twitter accounts from media from all those 12 countries, and we collected 1.7 million news tweets. Tweets, news tweets, right? So we took those tweets, all those tweets about COVID-19, obviously, we took those tweets and we collected the comments through those tweets, the replies and the quoted, the quoted retweets. That was in total 18.9 million comments from 3 million different users. So it's a lot of people. And this what called here shows from where we took more news tweets. So CNN, BBC, Sky News, The Washington Post, MSNBC, The New York Times, Fox News, NBC, CBS, Times of India, Toronto Star, The Irish Times. So there are media from many countries. Obviously, the media that we collected more news tweets are from the US and from the UK, because those are the media that tweets the most, mainly.

33:41

And we found a set of topics in those 1.8 million new streets. So what are the media talking about? The media are talking about cases and deaths, economic impact, educational impact, people's stories, authorities and politics, preventive measures, viruses spreading, vaccines and vaccination. And here we have some examples. I don't know. Those are examples of comments to news tweets like this, for example. This is a comment to a tweet about preventive measures. How to hold a kid's birthday party while social distancing. It's a person that is concerned about that. Another one. COVID-infected Donald Trump instantly refuses to take part in virtual debates with Joe Biden. Maybe some of you remember of that. I guess that was 2020. It's from a media in UK and it's that thing is altering politics. Sorry. I said that those are comments. No, those are the contents of the news tweet, right? Those are contents of news tweets.

35:13

And here it's a nice graph that we can see how the news tweets, sorry, how the comments are distributed over the time so most of the comments to new streets about covenants uh happened in 2020 mainly in April May June July there the months where the COVID-19 pandemic was being the center of the communications all over the world. And also we can see some of the keywords that were more used over the time. As you can see, COVID, which is dark blue, is more or less stable over time, while vaccine, which is the dark green, was just a little bit spoken in the beginning of the pandemic, but after 2021, after the first vaccinated person, it was one of their most commented topics, not topics, one of their keywords most used in comments right as you can see this another graph shows the distribution of emotions there and rather than

n that the prevalence of the emotions in the convenient in new streets over time in t
he prevalence of emotions tweets over time and the prevalence of emotions in users'
responses, their comments, to the COVID-19 news tweets over time. So the first grap
h shows the emotions of the news tweets and the second shows the emotions of the
comments.

37:20

As we can see, sadness is the most prevalent emotion in news tweets, followed by op
timism and anger and joy is the least prevalent by far, the most prevalent emotion in
user comments in all periods of time, followed by sadness, optimism and joy. So, as y
ou can see, there is a difference from how the news is framed and how people reacte
d to that. News are more sad and the comments are more angry to those news. It's a
n interesting result.

38:16

Those are the two graphs, shows the distribution of emotions in news to streets per
country and the distribution of emotions in comments to the new streets per country
. So sorry, the first one is the distribution of topics, not emotions. So the topic that is
most prevalent is preventive measures and is more prevalent in New Zealand and th
e UK and less prevalent in the Philippines and the US. As you can see the second one
, second one, authorities and politics is very prevalent in Australia. Australia talks a lo
t about authorities and politics. There is also this green one, virus spreading. It's very
prevalent in Nigeria. And those are something that pops through our eyes. And abo
ut the emotions and comments, USA and Australia are the countries with more ange
r in comments. Philippines is the country with less anger in comments. Also, Philippin
es have the most optimism in comments post optimism in comments and also more
user comments that there is no way to identify an emotion. Maybe because although
many tweets in Philippines are in English, maybe they use slangs, they use some lang
uage, which is particular from there.

40:08

So general highlights, we saw that sadness was the most observed emotion, mainly i
n news tweets, but also in people's stories, cases and deaths. So if we see this table ri
ght here, we can see that, we can see the team of the new street from cases and dea
ths until vaccines and vaccination. And we have this thing that we call the emotion lif

t, which is how many user comments we have more than average or less than average when the news topic is that. So, for instance, if the news topic is cases and deaths, we have less comments with anger and more comments with sadness than average. It's interesting to see. So sadness was the most observed emotion if we see people's stories. So there are much more people reacting with sadness if theme of the news tweet is people stories and the same is with kids and deaths and just that. And also it's interesting for example joy happens much more in comments if the news tweet is about educational impact. So maybe some, some stories that are interesting to people to see. We have much less sadness if theme of the news is about authoritarian politics. So people are not very sad when they read about authoritarian politics. So people are not very sad when they read about authoritarian politics, but they are more angry. So they are more angry than sad. Also, it's not in this table here, but American media had more response with anger. Philippines have more with optimism, which is what we saw in the last slide.

42:31

And that's that. We have a link here for our paper. It's open access, so you can read with no problem. And I'd like to thank my advisor, Professor Jaime Sikman from Brazil, Dr. Davos for inviting me to this speech, Professor Thiago Pardo, also a Brazilian professor, which I took many of the content of this presentation from his lectures. Also, I'd like to thank Australian National Academy of Science and EGPN for funding this research.

43:23

This is the presentation. I hope you like it and I'm open to questions anytime you can find me and I can be useful to solve some questions. Thank you. Have a good day, good night, good evening and bye.